Richness thresholds of $\mathtt{wmin}_{p/q}(\mathtt{rep}_{p/q}(1))$

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We compute the first richness thresholds of every minimal words of the form $\mathtt{wmin}_{p/q}(\mathtt{rep}_{p/q}(1))$ for every coprime integers p and q such that $2 \leq q < p \leq 9$. We compare them to:

- the richness thresholds of the q-ary expansions of $\sqrt{2}$ and π (which are strongly believed to be normal),
- the richness threshold of a random q-ary word,
- the asymptotic expected value for the richness threshold of a random q-ary word: $l \mapsto q^l \log(q^l)$.

The results are gathered in the next seven tables (one for each alphabet size $q = 2, 3, \ldots, 8$). These tables are to be read as follows: at the intersection of the row representing the word w and the column representing the length l:

- if the corresponding entry is positive, it is the richness threshold $\mathtt{rt}_w(l)$;
- if the entry is negative, its absolute value indicates how many words of length l are missing in the prefix of length 10^6 of w.

The last row gives the (asymptotic) expected value of the richness threshold for a q-ary random word.

l	1	2	3	4	5	6	7	8	9	10	11
$\texttt{wmin}_{3/2}(2)$	2	6	51	54	123	358	787	1479	2643	7272	18200
$\texttt{wmin}_{5/2}(2)$	3	6	11	52	221	228	661	992	2589	6507	16605
$\texttt{wmin}_{7/2}(2)$	2	8	34	86	115	201	905	1126	3160	5725	21722
$\texttt{wmin}_{9/2}(2)$	4	7	29	42	128	188	626	2365	5589	6548	23435
$\texttt{rep}_2(\sqrt{2})$	2	10	19	22	133	459	517	1806	3259	7185	18928
$\texttt{rep}_2(\pi)$	3	5	20	25	102	400	540	1351	3790	8034	17225
random word	3	6	17	68	171	185	548	1683	2989	6813	12979
$\lfloor 2^l \log(2^l) \rfloor$	1	5	16	44	110	266	621	1419	3194	7097	15615

l	12	13	14	15	16	17
$\texttt{wmin}_{3/2}(2)$	39358	65137	154725	390091	821322	-63
$\texttt{wmin}_{5/2}(2)$	31442	71030	189740	309169	827260	-64
$\texttt{wmin}_{7/2}(2)$	41938	77728	208773	384796	894414	-60
$\texttt{wmin}_{9/2}(2)$	32075	81088	190265	358020	914320	-61
$\texttt{rep}_2(\sqrt{2})$	32231	83298	166437	396117	847032	-53
$\texttt{rep}_2(\pi)$	35851	71909	160119	405148	824328	-63
random word	28729	78115	145390	454016	723874	-68
$\lfloor 2^l \log(2^l) \rfloor$	34069	73817	158991	340695	726817	1544487

Table 1: Richness thresholds for $q = 2$ (due to space constraints, the table is divided into two parts).

l	1	2	3	4	5	6	7	8	9	10	11
$\texttt{wmin}_{4/3}(3)$	3	16	165	389	1329	4607	21521	82002	198800	636034	-625
$\texttt{wmin}_{5/3}(3)$	5	32	70	396	1926	5768	16366	58164	252503	643016	-586
$\texttt{wmin}_{7/3}(3)$	4	19	98	573	1837	5099	16181	58426	169456	850881	-669
$\texttt{wmin}_{8/3}(3)$	3	35	79	342	1469	5752	17148	48774	224920	624652	-625
$\texttt{rep}_3(\sqrt{2})$	4	15	66	377	1290	7404	16511	56260	211187	790264	-629
$\texttt{rep}_3(\pi)$	6	15	119	348	1978	6379	15779	79122	183178	584098	-647
rand word	9	22	175	490	1118	5479	17382	66200	213250	692671	-616
$\lfloor 3^l \log(3^l) \rfloor$	3	19	88	355	1334	4805	16818	57663	194615	648719	2140774

Table 2: Richness thresholds for $q = 3$.

l	1	2	3	4	5	6	7	8	9
$\texttt{wmin}_{5/4}(4)$	4	62	333	1371	6932	33260	143470	826461	-5840
$\texttt{wmin}_{7/4}(4)$	4	47	430	2201	6680	31757	164198	902744	-5664
$\texttt{wmin}_{9/4}(4)$	10	39	309	1290	6417	35636	181371	857616	-5803
$\texttt{rep}_4(\sqrt{2})$	10	46	236	1486	8795	35655	149755	673039	-5818
$\texttt{rep}_4(\pi)$	4	55	236	1624	9359	34933	177634	702834	-5740
random word	9	35	268	2309	6858	36779	164101	604566	-5830
$\lfloor 4^l \log(4^l) \rfloor$	5	44	266	1419	7097	34069	158991	726817	3270678

Table 3: Richness thresholds for $q = 4$.

l	1	2	3	4	5	6	7	8
$\texttt{wmin}_{6/5}(\texttt{5})$	5	81	791	3939	26288	136085	942627	-30081
$\texttt{wmin}_{7/5}(\texttt{5})$	7	62	887	4374	37118	145118	916558	-29994
$\texttt{wmin}_{8/5}(\texttt{5})$	18	94	923	3629	23224	188051	-1	-30304
$\texttt{wmin}_{9/5}(\texttt{5})$	5	135	617	4571	20674	191759	752732	-30131
$\texttt{rep}_5(\sqrt{2})$	6	109	640	3435	22803	140840	844882	-30422
$\texttt{rep}_5(\pi)$	9	63	887	6655	24784	150127	-1	-30251
random word	10	109	472	4375	32282	171534	900053	-30399
$\lfloor 5^l \log(5^l) \rfloor$	8	80	603	4023	25147	150884	880161	5029493

Table 4: Richness thresholds for $q = 5$.

l	1	2	3	4	5	6	7
$\texttt{wmin}_{7/6}(\texttt{6})$	6	228	1316	7943	70475	518489	-7970
$\texttt{rep}_6(\sqrt{2})$	12	157	1150	8021	74028	468743	-7828
$\texttt{rep}_6(\pi)$	15	122	1534	12856	73806	583632	-7827
random word	15	137	1354	9120	61776	545873	-7908
$\lfloor 6^l \log(6^l) \rfloor$	10	129	1161	9288	69663	501577	3511045

Table 5: Richness thresholds for $q = 6$.

l	1	2	3	4	5	6
$\texttt{wmin}_{8/7}(\texttt{7})$	7	252	1921	18438	166562	-17
$\texttt{wmin}_{9/7}(\texttt{7})$	26	175	1765	16825	163228	-19
$\texttt{rep}_7(\sqrt{2})$	8	133	1979	17959	150677	-19
$\texttt{rep}_7(\pi)$	11	347	2119	17795	137191	-25
random word	13	245	1879	17087	185739	-33
$\lfloor 7^l \log(7^l) \rfloor$	13	190	2002	18688	163524	1373606

Table 6: Richness thresholds for $q = 7$.

l	1	2	3	4	5	6
$\texttt{wmin}_{9/8}(\texttt{8})$	8	405	2968	34776	303176	-5738
$\texttt{rep}_8(\sqrt{2})$	31	202	3321	27763	399910	-5742
$\texttt{rep}_8(\pi)$	15	369	2554	36141	355165	-5758
random word	21	375	3825	36054	346188	-5816
$\lfloor 8^l \log(8^l) \rfloor$	16	266	3194	34069	340695	3270678

Table 7: Richness thresholds for $q = 8$.